THREE DESCRIPTIONS OF THE DIMENSION OF AN ALGEBRAIC SPACE AGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original definition 04N6: sources/stacks-chow/tags.txt, source lines 6067–6067. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04MU: sources/stacks-chow/tags.txt, source lines 6055–6055. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following quantities are equal:*

(1) *The dimension of X.*
(2) *The supremum of the dimensions of the local rings of X.*
(3) *The supremum of $\dim_x(X)$ for $x \in |X|$.*

Proof. The numbers in (1) and (3) are equal by Definition 04N6. Let $U \to X$ be a surjective étale morphism from a scheme U. The supremum of $\dim_x(X)$ for $x \in |X|$ is the same as the supremum of $\dim_u(U)$ for points u of U by definition. This is the same as the supremum of $\dim(\mathcal{O}_{U,u})$ by Properties, Lemma 04MU. This in turn is the same as (2) by definition. □

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.